PORTFOLIO RISK ANALYTICS

EARLY WARNING & WATCHLIST REVIEW

Data Cleaning
Validation & Integrity

Anomaly Detection
Early Risk Identification

Predictive Modelling
Forward-Looking Analytics

Portfolio Risk
Comprehensive Assessment

Presented By
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PORTFOLIO RISK ASSESSMENT

Business Context & Problem Statement

BUSINESS CONTEXT

The bank has extended a **₦20 million** credit facility to a cafe and seeks to evaluate the borrower's financial performance and overall credit risk position.

KEY QUESTIONS

01. Did café sales genuinely decline in H2, or was the drop caused by data-quality issues?

02. Is the café meeting its financial loan covenants?

03. Should the café remain Normal, move to Watch, or be considered for Restructure?

CORE BUSINESS QUESTION

Should the bank continue to monitor the café under its normal portfolio process, or do the observed financial and credit-risk indicators justify placing the ₦20 million exposure on a watchlist?

PROJECT SCOPE & OBJECTIVES

PROJECT OBJECTIVES

01. Assess Borrower Performance
Sales and Financial Health

02. Identify Early Warning Signs
Revenue and transaction anomalies

03. Evaluate Financial Covenants
Key financial Ratios

04. Benchmark Credit Risk
Historical Borrower comparison

05. Evaluate Predictive Models
Credit risk classification

06. Recommend Portfolio Action
Monitoring and Watchlist decisions

SALES PERFORMANCE

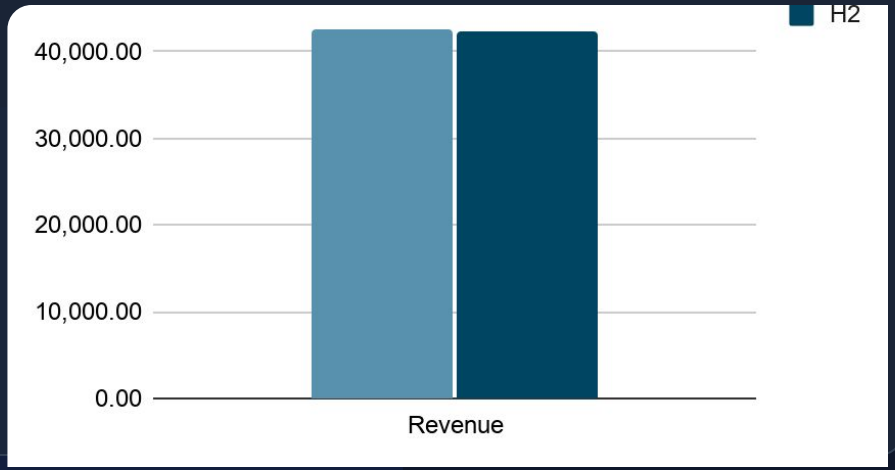
H1 vs H2 Revenue Analysis

H1 Revenue
42.55K

H2 Revenue
42.24K

Revenue Change
-0.73%

Revenue Distribution



Key Findings

Cafe revenue declined marginally by **0.73%** in H2 compared with H1.

This indicates broadly stable sales performance across periods rather than a significant operational decline.

DATA QUALITY

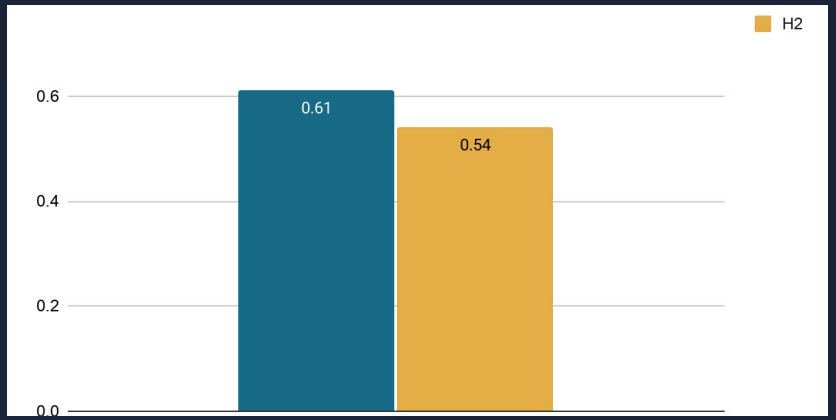
Data Quality: Is the Decline Genuine?

H1 Unrecoverable Records
0.61%

H2 Unrecoverable Records
0.54%

Data Quality Gap
0.07 pp

Unrecoverable Share Comparison



Key Findings

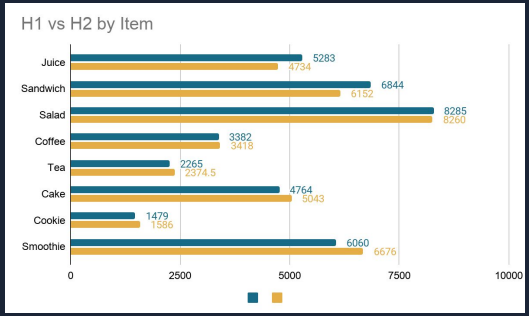
H2 had a slightly lower unrecoverable-record share than H1. The data-quality gap was only **0.07 percentage points**, well below the 5-percentage-point threshold.

Therefore, missing or unrecoverable records do not explain the H2 sales movement.

DETAILED BREAKDOWN

Sales Drivers: Items, Payment Methods & Service Style

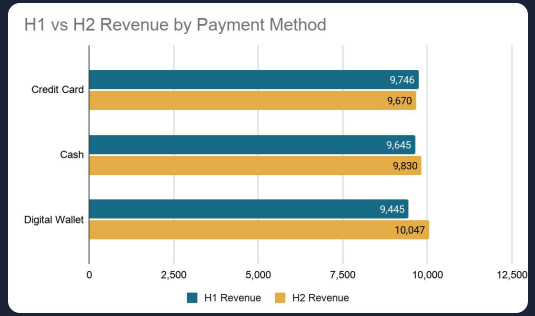
By Item



Juice **(-10.39%)** and Sandwich **(-10.11%)** recorded the largest revenue declines.

Smoothie **(+10.17%)** achieved the strongest revenue growth.

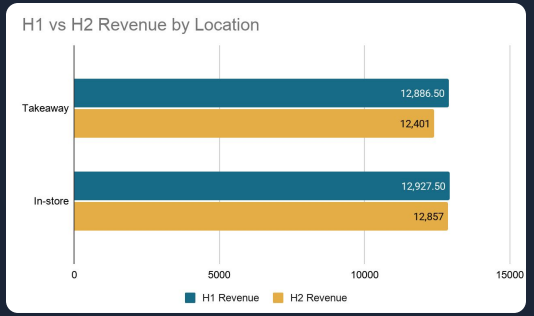
By Payment Method



Credit card revenue declined slightly by **0.78%**, while Cash & Digital Wallet grew.

Digital Wallet recorded the strongest increase at **6.37%**.

By Service Style



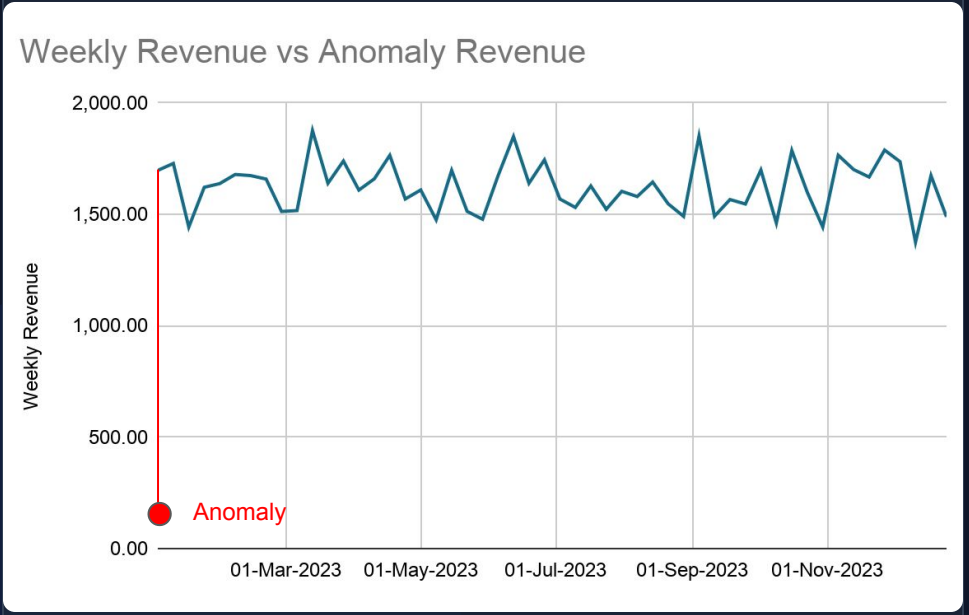
Both locations recorded lower H2 revenue.

Takeaway experienced a larger decline at **3.77%**, vs **0.55%** for in-store sales.

WEEKLY TRENDS

Weekly Trends & Anomalies

Weekly Revenue vs Anomaly Revenue



Anomaly Assessment

176 26 December 2022

One unusually low weekly revenue observation was flagged. It represents a partial opening week with only 20 transactions.

Bank Monitoring Implication

The cafe's weekly revenue should be monitored to see if there are unusual drops or a continuous decline that could affect its ability to repay the loan.

COVENANT ASSESSMENT

Financial Covenant Assessment

Net Profit Margin **PASS**

16.46%

Threshold: ≥10%

Interest Cover **PASS**

62.76

Threshold: ≥ 2.0

Current Ratio **PASS**

7.49

Threshold: ≥ 1.20

Year	Net Profit Margin	Current Ratio	Interest Coverage
2018	17.45%	5.17	49.07
2019	22.87%	14.64	70.5
2020	**16.46%**	**7.49**	**62.76**

KEY INSIGHT

All three financial covenants were met throughout the period reviewed. The café maintained strong profitability, liquidity and ability to cover interest expenses.

PORTFOLIO OVERVIEW

Historical Borrower Risk Patterns

Borrowers Analysed	Avg Age	Avg Credit Score	Avg Loan Amount
20,000	43 yrs	648.54	₦ 39,317.75

Avg Payment Delays (6m)	Avg Debt-to-Income	Avg Credit Utilization	Avg Existing Debt
1.49	67.39%	28.54%	₦ 52,668.68

Key Findings

The portfolio has an average debt-to-income ratio of **67.39%** and **1.49** payment delays over six months, indicating potential repayment pressure.

Data Limitation

All **20,000** records are classified as defaults, with PD values ranging from **99.67% to 100%**. The absence of non-default observations limits meaningful comparison and predictive model validation. Hence, the above descriptive risk analysis was used.

MODEL EVALUATION

Predictive Model Performance

Assessment of model suitability and limitations of the available loan book data

Total Borrowers	Unique Default Class	PD Range
20,000	1	99.67%–100%

Model Limitations

- All 20,000 borrowers are classified as defaulters.
- No non-default records are available for comparison.
- Model training and performance evaluation (ROC-AUC and default capture rate) cannot be reliably performed.

CONCLUSION

The loan book data must be corrected to include both default and non-default outcomes before a reliable predictive model can be developed and evaluated.

RISK ASSESSMENT

Café Risk Assessment & Limitation

Assessment of the café's repayment risk based on sales performance, financial covenants and available borrower information

Sales Performance

-0.73%

H2 revenue declined slightly against H1

Data Quality

H2 data quality was not worse than H1

Unrecoverable data was lower in H2 (0.54%) than H1 (0.61%).

Financial Covenants

3/3 Pass

All three financial covenants passed

Bank Classification Criteria

- **Normal:** No covenant breaches and PD below 20%.
- **Watch:** One covenant breach or PD between 20% and 39%.
- **Restructure:** Two or more covenant breaches or PD of 40% and above

⚠ Café's Current Position: Pending Final Classification

The café has passed all three reported financial covenants, with a slight H2 revenue decline of 0.73%. However, a reliable café-specific PD is unavailable. Therefore, a final classification under the bank's criteria cannot yet be established.

RECOMMENDATIONS & NEXT STEPS

Portfolio Recommendations & Action Plan

Recommended actions to strengthen credit risk monitoring and manage the café's repayment risk

01. OWNER ENGAGEMENT

Meet with the café owner to investigate the **0.73% revenue decline** and assess its impact on the **₦20 million** loan repayment capacity.

02. FINANCIAL AUDIT

Obtain complete financial statements and supporting expense records to verify covenant compliance and assess the café's financial position.

03. CASH FLOW REVIEW

Review the café's cash flow against its loan repayment schedule to determine its ability to meet upcoming repayments and identify any likely shortfall.

MONITORING & FOLLOW-UP FRAMEWORK

Normal - Relax — Maintain routine monitoring where repayment performance is satisfactory and no major risk indicators are present.

Watch - Hold — Maintain the existing credit exposure while increasing monitoring and following up on emerging risks.

Restructure - Tighten — Intensify monitoring, restrict additional credit exposure and assess the need for loan restructuring.

CAFE SALES HEALTH ANALYSIS

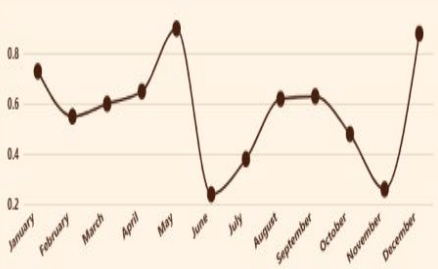


YEAR: All

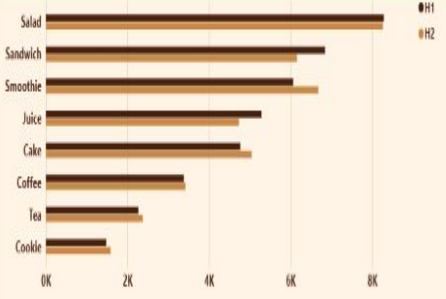
H1 Revenue	H2 Revenue	Number of Anomaly Weeks	HoH % Change
42.55K	42.24K	1	-0.73%



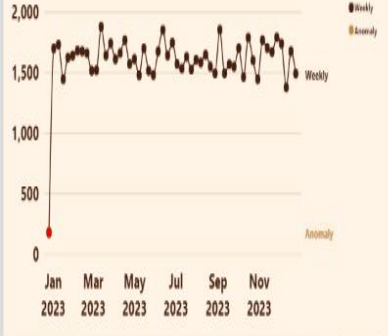
Monthly Unrecoverable / Revenue Share



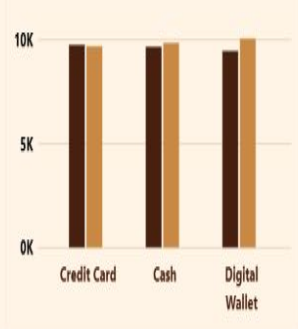
H1 vs H2 Revenue by Item



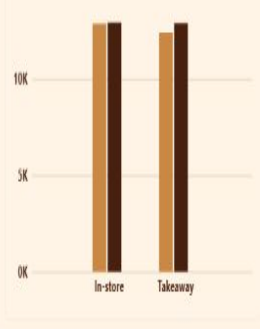
Weekly Revenue vs Anomaly Revenue



H1 vs H2 Revenue by Payment Method



H1 vs H2 Revenue by Location



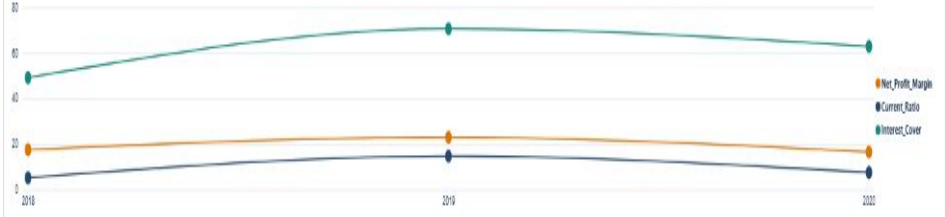
CONVENANT & WATCHLIST

CREDIT RISK MONITORING DASHBOARD

RISK CATEGORY: Normal | Restructure | Watch

Net Profit Margin	Interest Cover	Current Ratio	OVERALL COVENANT STATUS
16.46	62.76	7.49	Pass
STATUS: Pass	STATUS: Pass	STATUS: Pass	

Covenant Trend Analysis



Account Classification

AccountKey	Account	SubAccount	Account Category
10	Cash & Cash Equivalents	Cash at Bank	Current Asset
20	Cash & Cash Equivalents	Cash in Hand	Current Asset
30	Receivables	Trade Receivables	Current Asset
40	Receivables	Other Receivables	Current Asset
41	Receivables	Interest Receivable	Current Asset
42	Receivables	Dividend Receivable	Current Asset
50	Receivables	Allowance for Bad Debt	Current Asset
60	Inventory	Inventory	Current Asset
70	Other Current Assets	Prepaid Expenses	Current Asset
75	Investments	Securities	Current Asset
110	Trade Payables	Trade Payables	Current Liability
120	Other Payables	Accrued Expenses	Current Liability

Loan Details

Loan_ID	Sum of Credit_Score	Sum of Payment_Delays_6mo	Sum of Debt_to_Income_Ratio	Risk_Category
6b647457-1cfc-2491-b840-1eea5f6f0530	300	4	0.11	
6be7c689-c31e-4b1b-adcc-6e67ebd10c91	300	0	0.19	
c1958d2c-ee8a-4d7e-80f1-d61410160ee4	300	3	0.67	
ff2d7bb0-0ccc-45bf-b87e-c320061619b0	300	1	0.69	
00f999de-fca3-49e8-b2fb-f31264ecfbeb	310	3	0.05	
37b85ad7-a621-4c82-84ec-df5a190850b1	311	1	0.60	
559024f9-da9d-46dc-8ea6-eee3404d3597	314	0	0.70	
a294f41e-7e14-488b-aa3b-6003df54c80e	321	1	0.73	
cd11431d-4311-41ae-bbb4-87f7e3f60be5	324	0	0.42	
bfbfcbda-ee57-451c-6dcb-1cb149e7102d	327	3	0.40	
Total	12970735	29775	13,478.11	

To: Relationship Manager
From: Mary Ojolabi
Subject: Early Warning Review
Date: 6 November 2022

EARLY WARNING REVIEW MEMORANDUM

This memo presents the findings of the early warning review conducted to assess the cafe's sales performance, evaluate compliance of the company against its financial covenants and examine potential borrower risk indicators in order to determine the cafe's ability to meet its financial and credit obligation.

1.CAFE SALES PERFORMANCE

The first part of the analysis examined whether the decline in the cafe's sales during the second half of the year (H2) reflected an actual reduction in business performance or was influenced by incomplete and inaccurate transaction records.

Data Quality Assessment & Impact

Of 10,000 transaction records reviewed, 1,398 were recovered during data cleaning, while 58 remained unrecoverable. Unrecoverable records represented 0.61% and 0.54%of H1 and H2 transactions. This reduction suggests that data quality issues did not materially account for the recorded 73% decline in revenue.

Weekly Revenue Anomaly

One significant weekly revenue anomaly was identified in the week beginning 28 December 2022, with a z-score of -6.19, indicating an unusually low revenue level.

2.FINANCIAL COVENANT ASSESSMENT

The review assessed the company's Net Profit Margin, Current Ratio and Interest Cover against the specified covenant thresholds. The initial SQL results indicated compliance, however, incomplete and missing interest expense records in 2019 and 2020 limit the reliability of the results.

Covenant	Threshold	2018	2019	2020
Net Profit Margin	≥10%	17.45%	22.87%	18.46%
Current Ratio	≥1.20	5.17	14.64	7.49
Interest Cover	≥2.0	49.07	70.5	62.76

Limitation: Incomplete expense records overstates Net profit margin, while absence of recorded interest expense in 2019 and 2020 prevents reliable interest cover calculations.

3.CREDIT RISK ASSESSMENT

Analysis of the 20,000 loan records examined repayment delays, credit scores and debt-to-income ratios as potential risk indicators. However, all records showed a default status of 1 (100%), making reliable default prediction and model validation impossible. The review was therefore limited to descriptive risk analysis.

4.OVERALL RISK ASSESSMENT

The 73% revenue decline is a significant warning sign which requires closer monitoring. However, incomplete financial records and the absence of a reliable default prediction limit the ability to assign a definitive watchlist status. Further financial validation is recommended before final classification.

5.RECOMMENDED ACTIONS - NEXT 30 DAYS

- Meet with the cafe owner to investigate the 0.73% revenue decline and assess its impact on the ₦20 million loan repayment capacity.
- Obtain complete financial statements and supporting expense records to verify covenant compliance and assess the cafe's financial position.
- Review the cafe's cash flow against its loan repayment schedule to determine whether it can meet upcoming repayments and identify any likely shortfall

6. MONITORING RULES

- **Normal - Relax**: Maintain routine monitoring where repayment performance is satisfactory and no major risk indicators are present.
- **Watch - Hold**: Maintain the existing credit exposure while increasing monitoring and following up on emerging risks.
- **Restructure - Tighten**: Intensify monitoring, restrict additional credit exposure and assess the need for loan restructuring.

Please implement the recommended actions within 30 days to address the identified risks and protect the bank's ₦20 million loan exposure.